Characterisation of interaction potentials for Nitrogen at low temperature ($T < 600\ K$) from transport properties and rotational relaxation

Domenico Bruno[1] and Aldo Frezzotti[2]

[1] CNR, Institute for Plasma Science and Technology, Bari, Italy
[2] Politecnico di Milano, Dipartimento di Scienze e Tecnologie Aerospaziali

HyperStructures 2026 - Continuum-Kinetic Models in Hypersonic Transitional Flows
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Summary

PES from Perugia

An Experimental Potential Energy Surface and Calculated Rotovibrational Levels of the Molecular $N_2 - N_2$ Nitrogen Dimer

V. Aquilanti *et al.*, *J. Chem. Phys.* **117**, 615 (2002)

- Motivation: characterization of the $N_2 - N_2$ dimer
- Approach: analysis of molecular beams scattering data and second virial coefficients
- Results: bound rotovibrational states of the dimer for $J \leqslant 6$

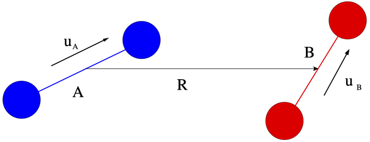
PES from Hellmann

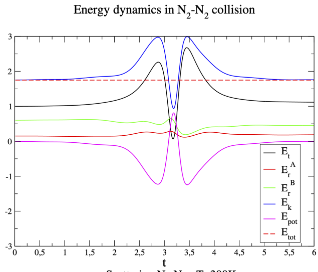
Ab initio potential energy surface for the nitrogen molecule pair and thermophysical properties of nitrogen gas

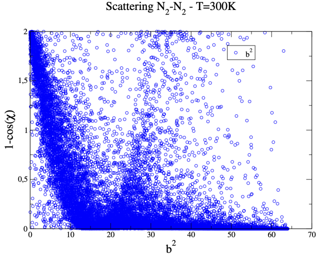
R. Hellmann, *Molecular Physics* **111**, 387 (2013)

- Motivation: Thermophysical properties (transport, virial coefficients)
- Approach: fitting over 400 energy configurations from quantum electronic structure calculations
- Results: shear viscosity thermal conductivity, 2^{nd} & 3^{rd} virial coefficients

Trajectory & Scattering simulations







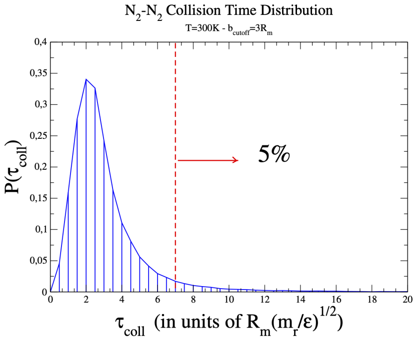
Equations of motion

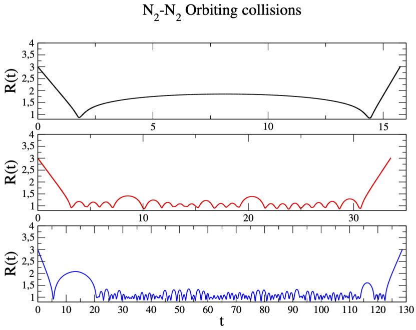
$$
\begin{aligned}
\dot{\boldsymbol{R}} &= \boldsymbol{v} \\
m_r \dot{\boldsymbol{v}} &= \boldsymbol{F}_B \\
\dot{\boldsymbol{u}}_A &= \boldsymbol{\omega}_A \wedge \boldsymbol{u}_A \\
\mathcal{I} \dot{\boldsymbol{\omega}}_A &= \boldsymbol{T}_A \\
\dot{\boldsymbol{u}}_B &= \boldsymbol{\omega}_B \wedge \boldsymbol{u}_B \\
\mathcal{I} \dot{\boldsymbol{\omega}}_B &= \boldsymbol{T}_B
\end{aligned}
$$

CT-MC estimation of N_2 transport

Approach

Monte Carlo integration of Chapman-Enskog expressions (1^{st} approximation)

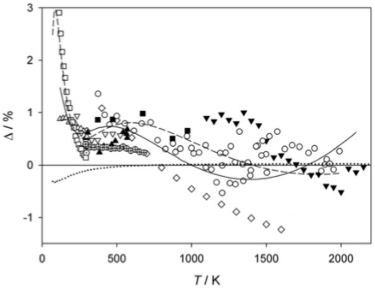


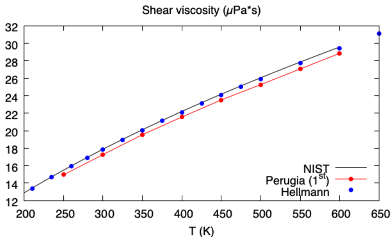


Results I. Shear viscosity

Theory: C-E 1^{st} approximation

$$\eta = \frac{5k_B T}{8\Omega^{(2,2)}} \qquad \Omega^{(2,2)} \sim \int (1 - \cos^2\chi) \ldots dgdb$$





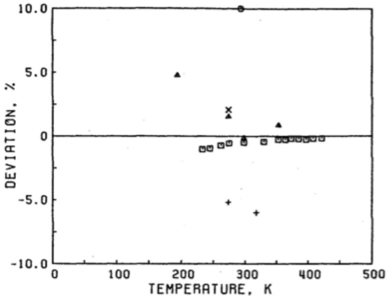
NIST

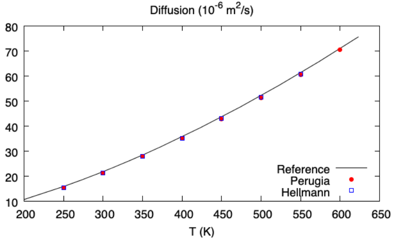
Viscosity and Thermal Conductivity Equations for Nitrogen, Oxygen, Argon, and Air
E.W. Lemmon, R.T. Jacobsen, *International Journal of Thermophysics* **25** 21 (2004)

Results II. Diffusion

Theory: C-E 1^{st} approximation

$$\mathfrak{D} = \frac{3k_B T}{8nm\Omega^{(1,1)}} \qquad \Omega^{(1,1)} \sim \int (1-\cos\chi)\dots dgdb$$





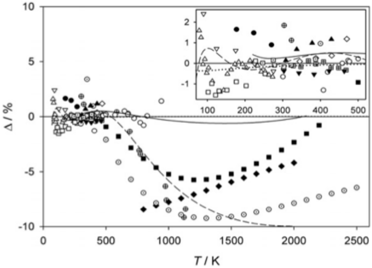
Source

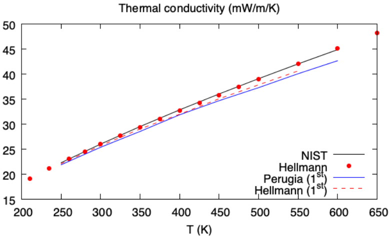
Equilibrium and Transport Properties of Eleven Polyatomic Gases At Low Density
A. Boushehri *et al.*, *Journal of Phys. Chem. Ref. Data* **16** 445 (1987)

Results III. Thermal conductivity

Theory: C-E 1^{st} approximation

$$\lambda_1 = \frac{5}{2}\eta c_{tr}\left[1 - \frac{c_{rot}}{c_{tr}}\frac{5/2\eta - \rho\mathfrak{D}}{2p\tau}\right] \qquad \lambda_2 = \rho\mathfrak{D}c_{rot}\left[1 + \frac{5/2\eta - \rho\mathfrak{D}}{2p\tau}\right]$$



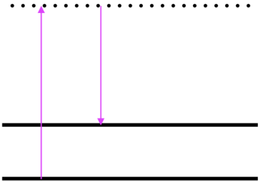


Source

Viscosity and Thermal Conductivity Equations for Nitrogen, Oxygen, Argon, and Air
E.W. Lemmon, R.T. Jacobsen, *International Journal of Thermophysics* **25** 21 (2004)

Raman scattering

Line mixing in the isotropic fundamental Q branch



$$\Gamma_j = -\frac{1}{2\pi c_0} \sum_{i \neq j} W_{ij} \qquad (1)$$

$$K_{ij} = 2\pi c_0 \, k_B T \, W_{ij} \qquad (2)$$

Sitz & Farrow

- $J = 0 - 14$
- $T = 298\ K$

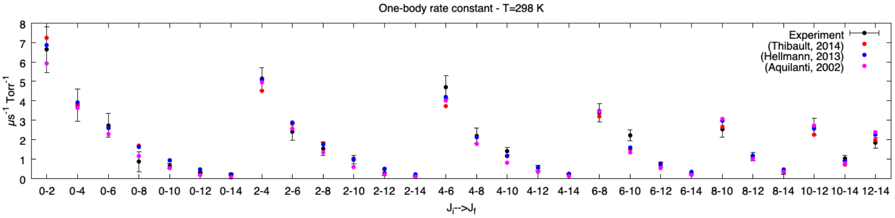
Calculation

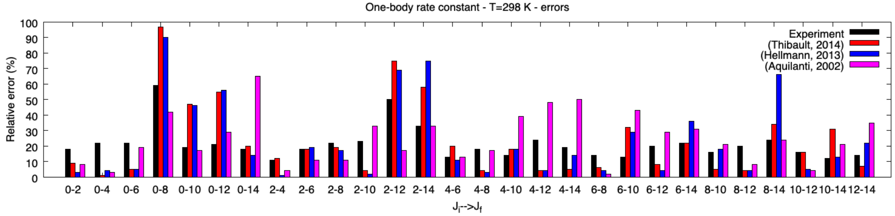
- $K_{ij} = \sum_k \frac{n_k}{n} \sum_l K_{ij}^{kl}$
- $J_{max} = 26$, $\Delta J_{max} = 16$

Pump-probe measurements of state-to-state rotational energy transfer rates in N_2 (v=1)
G.O. Sitz, R.L. Farrow, *J. Chem. Phys.* **93** 7883 (1990)

Raman scattering

Results for One-body rates at $T = 298\ K$





Line coupling effects in the isotropic Raman spectra of N_2: A quantum calculation at room temperature
F. Thibault, C. Boulet, Q. Ma, *J. Chem. Phys.* **140** 044303 (2014)

Rotational relaxation

Bulk viscosity

$$\eta_V = \frac{c_{int}}{c_v^2} p\tau = \frac{c_{int}}{c_v^2}\frac{\pi}{4}\eta Z = \left(\frac{c_{int}}{c_v}\right)^2 \frac{k_B T}{2\left[\left[(\Delta\epsilon)^2\right]\right]}$$

Rotational relaxation number (Parker 1959)

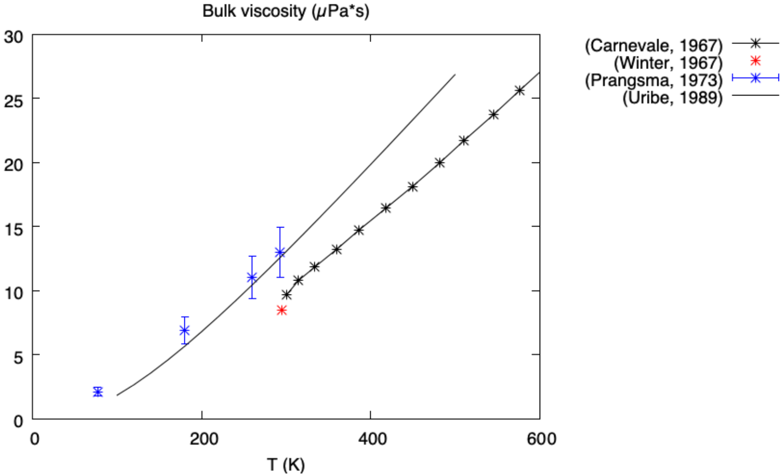
$$Z = Z_\infty \left[1 + \frac{\pi\sqrt{\pi}}{2}\sqrt{\frac{T_0}{T}} + \left(\frac{\pi^2}{4} + 2\right)\frac{T_0}{T} + \pi\sqrt{\pi}\left(\frac{T_0}{T}\right)^{\frac{3}{2}}\right]^{-1}$$

Sound absorption coefficient

$$\Gamma = \frac{1}{2}\left[\frac{1}{\rho}\left(\frac{4}{3}\eta + \eta_V\right) + (\gamma - 1)\frac{\lambda}{\rho c_p}\right]$$

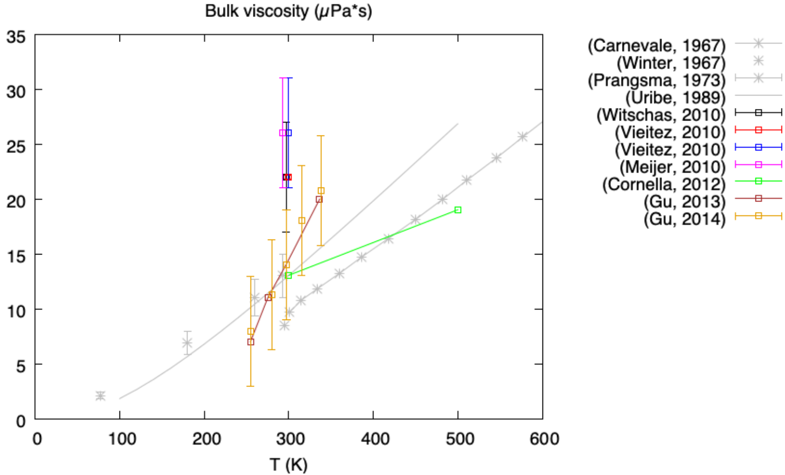
Rotational and Vibrational Relaxation in Diatomic Gases
J.G. Parker, *Physics of Fluids* **2** 449 (1959)

Literature results: ultrasound absorption



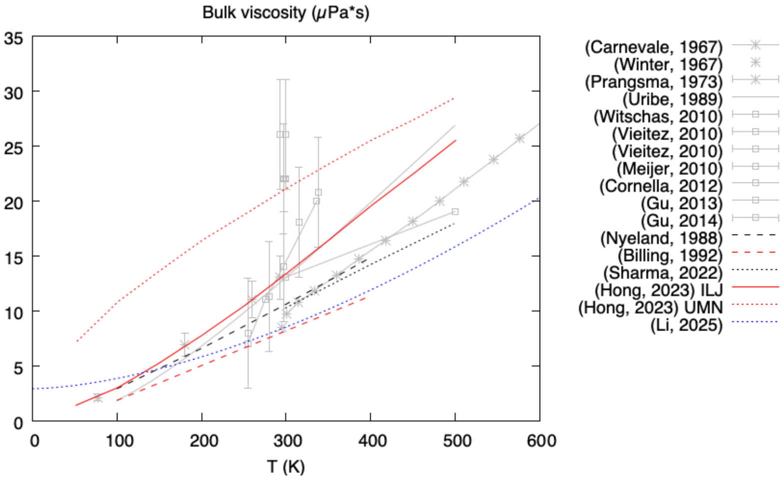
Thermal conductivity of nine polyatomic gases at low density
F. J. Uribe, E. A. Mason, J. Kestin, *Journal of Physical and Chemical Reference Data* **19** 1123 (1990).

Literature results: RBS/CRBS

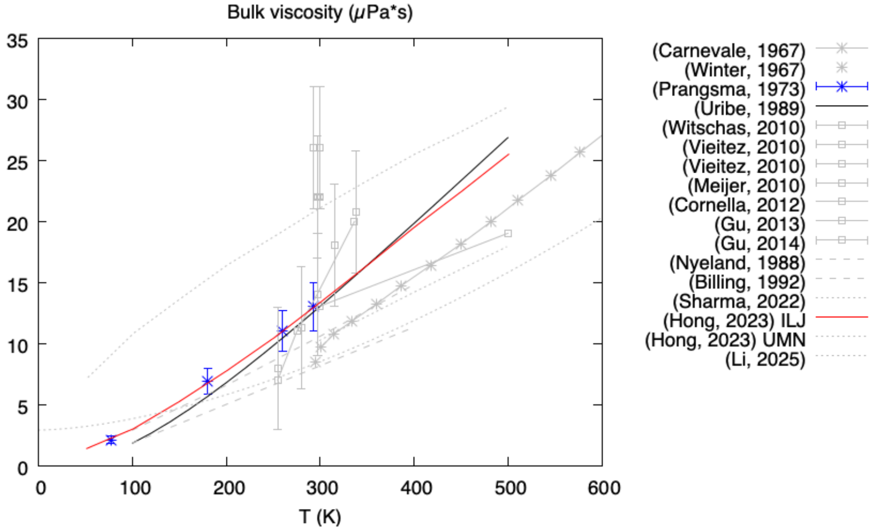


Finding the bulk viscosity of air from Rayleigh-Brillouin light scattering spectra
D. Bruno, A. Frezzotti, S.H. Jamali, W. van de Water, *J. Chem. Phys.* **158** 031101 (2023).

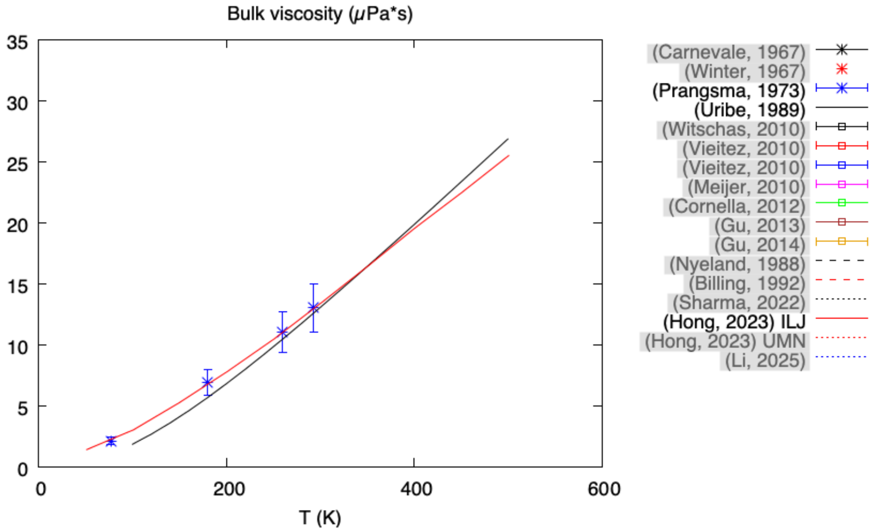
Literature results: PES



Literature results: reference



Literature results: reference



Bulk viscosity

Calculation approaches

- Monte Carlo evaluation of collision integrals (1^{st} C-E approximation)

  Classical theory of transport phenomena in dilute polyatomic gases
  N. Taxman, Phys. Rev. **110** *1235 (1958).*

- Homogenous relaxation

$$\eta_V = \frac{c_{int}}{c_v^2} p\tau$$

- Green-Kubo expression (Equilibrium simulation)

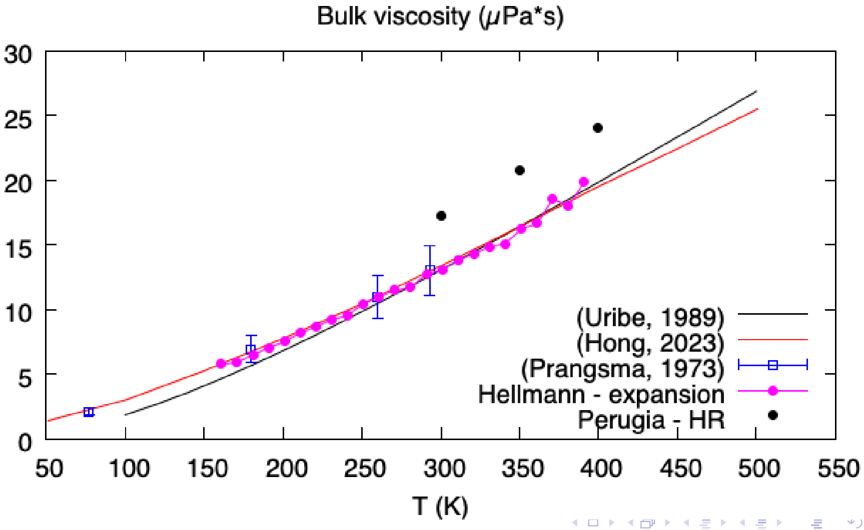
$$\frac{4}{3}\eta + \eta_V = \frac{1}{V k_B T} \int \langle \boldsymbol{P}(0) : \boldsymbol{P}(t) \rangle dt$$

- Adiabatic expansion (Non-eq simulation)

$$\frac{\partial f}{\partial t} - \frac{\delta}{3} \boldsymbol{v} \cdot \nabla_{\boldsymbol{v}} f = C(f, f)$$

Bulk viscosity

Results



Rotational relaxation

Role of inelastic collisions

OBR

$$K_{ij} = \sum_k \frac{n_k}{n} \sum_l K_{ij}^{kl}$$

$$J_{max} = 26$$
$$\Delta J_{max} = 16$$

Bulk viscosity

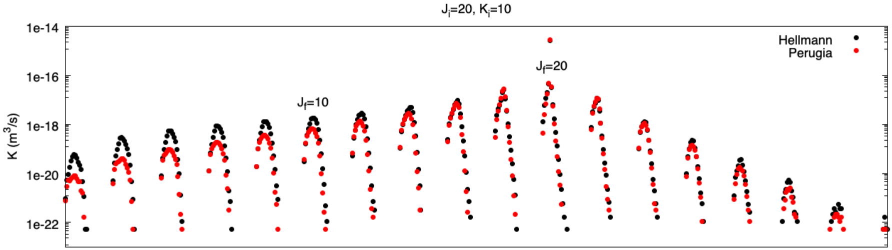
$$\eta_V = \left(\frac{c_{int}}{c_v}\right)^2 \frac{k_B T}{2\left[\left[(\Delta\epsilon)^2\right]\right]}$$

$$\left[\left[(\Delta\epsilon)^2\right]\right] = \frac{1}{8} \sum_{i,j,k,l} (\Delta\epsilon)^2 \frac{n_i}{n} \frac{n_j}{n} K_{ij}^{kl}$$

$$J_{max} = 30$$
$$\Delta J_{max} = 26$$

Results

Two-body rates



Conclusions

- Classical Trajectories are used to evaluate transport properties from PES
- Properties that depend mostly on isotropic interaction show good agreement with experiment: η, $\mathfrak{D}$, λ
- Rotational relaxation rates match the results of Raman experiments for low inelasticity
- Strongly inelastic transitions show important differences. These can be revealed from bulk viscosity measurements.

Outlook and future plans

Can we derive conclusions on the PES?

- Rotational relaxation is caused by the anisotropic part of the potential
- Look at the 3^{rd} PES (Cappelletti, 2008), used in Raman studies (Thibault, 2014) and vibrational kinetics (Hong, 2023)

Numerical estimation of N_2 bulk viscosity

- Objective: simulations of gas expansions with constant divergence δ.
- Theory: Physics can be cast as a space homogeneous problem. The local gas expansion, as seen by an observer moving with the bulk velocity is governed by the following Boltzmann equation:

$$\frac{\partial f}{\partial t} - \frac{\delta}{3}\boldsymbol{v} \cdot \nabla_{\boldsymbol{v}} f = C(f, f)$$

- $f(\boldsymbol{v}, \boldsymbol{\gamma}, t)$ is the phase averaged distribution function of molecular velocity $\boldsymbol{v}$ and angular momentum $\boldsymbol{\gamma}$.
- $C(f, f)$ describes binary collisions between linear rigid rotators, under the action of intermolecular forces derived by the PES.

A particle method

- CT-DSMC particle scheme more conveniently formulated from the equivalent form of the above BE:

$$\frac{\partial f}{\partial t} - \frac{\delta}{3}\nabla_{\boldsymbol{v}} \cdot (\boldsymbol{v} f) + \delta f = C(f, f)$$

- The third term on l.h.s. makes the density ρ vary as:

$$\rho(t) = \rho(0)\exp(-\delta t)$$

- The second term causes an isotropic velocity rescaling which increases or decreases T_t, depending on the sign of δ.
- The second term acts only on the translational degrees of freedom.
- The coupling with rotational degrees of freedom is provided by the collision term, $C(f, f)$.

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Rotational relaxation

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Bruno & Frezzotti

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Raman line broadening

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NIST

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